Calcaric Fluviosols

Description:

Calcaric Fluviosols are young, fertile soils formed on alluvial deposits along rivers, streams, and floodplains. The term "Calcaric" indicates the presence of a significant calcium carbonate content, often occurring as soft lime, nodules, or minor calcrete layers within the profile.

These soils are typically medium-textured (sandy loam to loam), light brown to yellowish, and have weakly developed horizons due to ongoing sediment deposition. Organic matter content is usually moderate near the surface, but decreases with depth. They are highly valued for agriculture due to their nutrient richness and good workability.

Drainage:

- **Wet season:** Generally well-drained on sloping alluvial terraces; poorly drained in depressions or low-lying floodplains where waterlogging can occur temporarily.
- **Dry season:** Soil dries quickly but retains enough moisture for crop growth due to moderate clay content and good aggregation.

Landscape:

Found along river valleys, floodplains, alluvial terraces, and lowland plains. Common in regions with seasonal flooding, such as along the Tana, Ewaso Ng'iro, and Athi rivers in Kenya.

Depth:

- Moderately deep to deep (50–150 cm), with unrestricted rooting depth where sediment deposition is continuous and free of coarse gravel or calcrete layers. Shallow areas may occur in older terraces with underlying calcareous material.

Workability:

- **When wet:** Friable and easy to till; responds well to standard cultivation methods.
- **When dry:** Maintains structure, easy to prepare seedbeds. Minimal crusting due to moderate clay and organic matter content.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Moderate to high (10–20 cmol(+)/kg), reflecting clay content and organic matter.
- **Base saturation:** High (>60%) due to abundant calcium carbonate and other basic cations.
- **Nutrients:** Adequate levels of calcium, magnesium, and potassium. Nitrogen may be limiting and requires supplementation. Phosphorus is generally moderate but can be enhanced with fertilization.
- **Organic matter:** Moderate at the surface (2–4%), decreasing with depth.
- **Micronutrients:** Generally sufficient; zinc or iron deficiencies may occasionally occur.

pH:
Slightly alkaline to neutral (6.5–8.0), depending on calcium carbonate content and local hydrology.

Management:

- Incorporate organic matter (manure, compost, green manure) to improve fertility and moisture retention.
- Apply nitrogen and phosphorus fertilizers to optimize crop yields.
- Use proper irrigation practices on floodplain areas to prevent leaching or waterlogging.
- Employ soil conservation measures (cover crops, grass strips) on sloping terraces to prevent erosion.
- Periodically test soil to monitor pH and nutrient status for optimal crop management.

Suitable crops:
Ideal for cereals (maize, rice, wheat), pulses (beans, cowpeas), vegetables (tomatoes, onions, cabbages), and fodder crops under irrigation or rainfed conditions, due to fertile, deep, and workable soil.